UPISC UNIVERSITY OF PITTSBURGH INFRASTRUCTURE SENSING

COLLABORATION WORKSHOP

NATIONAL ENERGY TECHNOLOGY LABORATORY

Load Rating Analysis of T-beam Bridges

Majid Eshraghi[1], Piervincenzo Rizzo[2], Fabio Matta[3]

[1] Ph.D. Student, Department of Civil and Environmental Engineering, University of Pittsburgh, Pittsburgh, PA 15261, USA.

[2] Professor, Department of Civil and Environmental Engineering, University of Pittsburgh, Pittsburgh, PA 15261, USA.

[3] Professor, Department of Civil and Environmental Engineering, University of South Carolina, Columbia (SC), USA.

INTRODUCTION

Load rating analysis (LRA) quantifies the safe live load carrying capacity of an existing bridge. When one or more structural components have an insufficient rating, limitations such as lane restriction, speed limit, weight posting, or closure, are imposed on the asset.

LRA identifies the bridge controlling structural component (one with the lowest rating factor (RF)), which determines the overall rating of bridge.

The **inventory rating** determines a live load that can be allowed indefinitely; the **operating rating** determines the maximum live load a bridge can carry.

A bridge with an RF ≥ 1.0 at both levels does not require posting. If operating-RF ≥ 1.0 and inventory-RF < 1.0, the bridge may not safely carry a state's legal loads, often requiring a weight limit. When RF < 1.0 at both levels, the bridge must be posted and further evaluated.

Three LRA methods exist: Allowable Stress Rating (ASR), Load Factor Rating (LFR), and Load and Resistance Factor Rating (LRFR).

LRA is governed by 2D analyses that inherently simplify the three-dimensional (3D) load distribution behavior of real structures, often yielding overly conservative RFs and unnecessary operational restrictions.

METHODOLOGIES

Demonstrate and quantify the conservatism of 2D analyses versus 3D accurate modeling for **three T-beam bridges**.

Create high-fidelity finite element (FE) models to perform a static analysis of these bridges under real-case scenario to recognize over-conservative postings.

Compare the FE results with the 2D-based RFs and distribution factors (DF) reported in the inspection reports of those bridges.

Determine the controlling failure mechanisms.

Evaluate the effect of damage on RFs and DFs.

METHODOLOGIES

Task 1: Create the FE models of three bridges under:

- as built scenario (based on standardized plans),
- current condition (based on available bridge inspection reports),
- additional damage with respect to the current condition,
- current condition with 20% lower concrete compressive strength and reinforcement yield stress.

Task 2: Structural analysis (for each of the above scenarios)

Task 3:

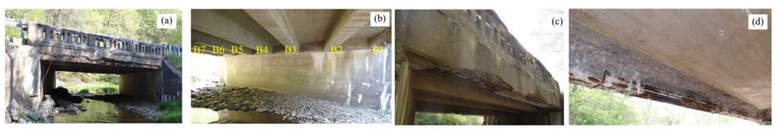
- Perform the LRA for the above scenarios and compare the results with the 2D LRA found in the latest inspection reports
- Verify punching shear
- Compute load distribution factors and compare with 2D values.



Harbor Creek Township Bridge



Sugar Creek Borough 1703 Bridge

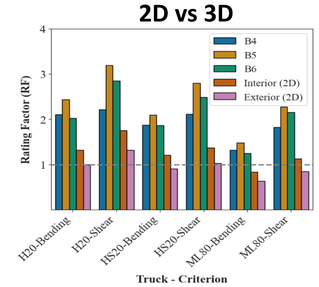


Sugar Grove Township Bridge

PRELIMINARY RESULTS & PERSPECTIVES

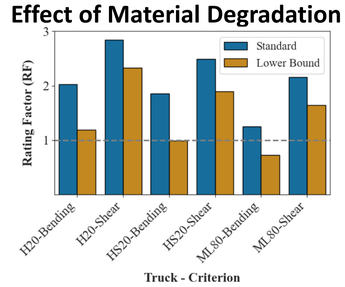
Sample Results for Sugar Groove Township Bridge:

2D vs 3D



More than 70% increase in RFs with respect to current estimation based on conventional 2D analysis.

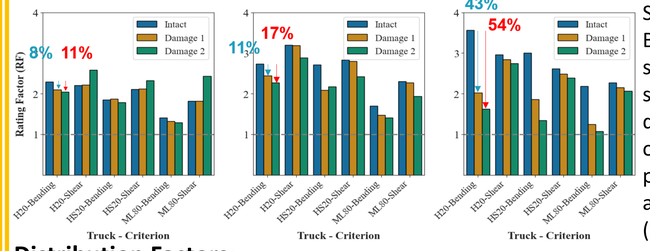
Effect of Material Degradation



33-57% decrease in bending & 13-24% decrease in shear RFs.

Effect of Damage on beams:

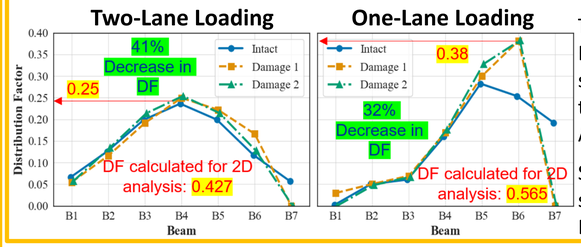
B4 (left) **B5 (center)** **B6 (right)**



Severe damage to B7 in *Damage 1* scenario led to significant decrease in RFs of other beams, particularly the adjacent beam (B6).

Distribution Factors:

Two-Lane Loading **One-Lane Loading**



The obtained DF from 3D FE modeling is significantly lower than the DFs calculated from AASHTO relations.

Severe damage to beams substantially changes the DFs.

ACKNOWLEDGEMENT

The authors gratefully acknowledge the support provided by the University of Pittsburgh's IRISE Consortium.